

ratings, that were achievable in larger form factors, are now being implemented in smaller surface-mount devices.

Additionally, in over-voltage protection, surge-handling capabilities of larger metal-oxide varistor products are now being achieved in smaller form factors, and even in surface-mount technologies.

A recent practice is to pack together sub-assemblies of a combination of two protection schemes. Natarajan M.M., vice president for South Asia of Arrow Asia-Pacific, explains, "To overcome electrical over-stress (EoS) and to protect equipment from consequent damage, an innovative solution is brought about by hybrid micro-assemblies, also called integrated product devices, in which two circuit protection devices, usually a combination of over voltage (OV) and over current (OC), are packaged together to provide efficient, cost-effective and space-saving circuit-protection solutions."

Patil adds, "A digital isolator is a unique ADI product that helps to form an isolation barrier between two sections of the circuit, namely, the low voltage section (very sensitive) and the other side that is facing the field (prone to damage by unwanted signals). A digital isolator prevents unwanted high-potential voltage from affecting the sensitive portion of circuit by creating a barrier. Another impressive technology is our Latch-Up Proof multiplexers (MUXes)."

He continues, "A latch-up is a condition where a low-impedance path is formed between supply and ground, causing excess current to flow, heating up the affected part and eventually burning it out. The MUX or the switch is typically the first element to get exposed in a signal/data-acquisition system, so one of the ways to protect the circuit is to make this MUX as robust as possible. We do this by using a circuit-design technique known as trench isolation."

Listen up, you

It has been mentioned umpteen times that circuit protection is extremely critical. Let us make that umpteen plus one, with this take by Tim Patel, technical marketing manager, Electronics Business Unit, Littelfuse, "Circuit protection is a critical part of design; it should not be an afterthought. You need to anticipate what faults can occur, what can cause reliability issues, what faults can come into the system that can cause surge events, how a short-circuit event occurs, and what will happen if you do not protect against it. You may need to redo the board layout, costing money and lost development time. You may end up with a less-than-optimal protection device or location, which results in functional failures, poor reliability and safety issues such as shock or fire."

He adds, "Let me tell you a story. Smartphone and tablet industries have evolved so much that chargers are now directly connected to the wall; the wire is just a USB cable. Any short-circuit


KORE KEY OPERATION & ELECTROCOMPONENTS PVT. LTD.
 Total LED Lighting Solutions
 AN ISO 9001:2008 CERTIFIED COMPANY









NEW Launch
LED Flood Light Fixtures
 10W/20W/50W/
 100W/150W/180W



LED High Bay Light Fixtures
 50W/120W/
 150W/180W



PHILIPS DRIVER
 9W, 10W, 12W, 20W

























- LEDs: High watt, SMD etc.
- LED Downlights / Fixtures.
- Modules for Street Lights & Bulbs.
- LED Bulbs-MR16, B22, GU10, E27.
- LED Modules, Rigid Strip, Strip Light.
- LED Street Lights/Bay Lights & Flood Light Fixtures.

OFFICE: Plot no-48, 1st Floor, Street No-3, Rajasthan Udyog Nagar,
 Near Jahangirpuri Metro Station, G T Karnal Road, Delhi-110033.
 Ph. No.: 011-45380073/47481516 Mob.: 9953557477
 Email: sales@keyoperation.in, Website : www.keyoperation.in

SHOWROOM: Stall No: 15, Opp. Shop No: 355, Old
 Lajpatrai Market, Deewan Hall Road.
 DELHI-110006.
 Ph. No. : 91 11 23861005